

Citi Bike Demand Proposal

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com> (<http://rmarkdown.rstudio.com>).

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Use control+Enter to run the code chunks on PC. Use command+Enter to run the code chunks on MAC.

Load Packages

In this section, we install and load the necessary packages.

Import Data

In this section, we import the necessary data for this lab.

Descriptive Analysis

[OBJECTIVE AND QUESTIONS HERE]

Explore your data

Explore the dataset using 5 functions: `dim()`, `str()`, `colnames()`, `head()` and `tail()`.

```
# Explore the dataset using 5 functions: dim(), str(), colnames(), head() and tail  
  
dim(citibike)
```

```
## [1] 31414    25
```

```
str(citibike)
```

```
## 'data.frame': 31414 obs. of 25 variables:
## $ DemandDate : chr "6/1/2017" "6/1/2017" "6/1/2017" "6/1/2017" ...
## $ StartStationId : int 83 128 128 152 161 168 173 223 229 239 ...
## $ EndStationId : int 3368 309 394 3329 518 402 526 168 309 421 ...
## $ DemandTime : chr "evening" "daytime" "daytime" "evening" ...
## $ Demand : int 20 37 37 28 79 53 47 30 48 41 ...
## $ Temperature : num 72.5 72.5 72.5 72.5 72.5 ...
## $ DailySnowFall : num 0 0 0 0 0 0 0 0 0 0 ...
## $ DailyPrecipitation : num NA NA NA NA NA NA NA NA NA NA ...
## $ Customer_prop : num 0.0965 0.0965 0.0965 0.0965 0.0965 ...
## $ DayOfWeek : chr "Thursday" "Thursday" "Thursday" "Thursday" ...
## $ Month : int 6 6 6 6 6 6 6 6 6 6 ...
## $ StartPerCapitaIncome : int 61845 115888 115888 166343 79481 113379 90151 113379 851
99 71656 ...
## $ StartPctHouseholdsNoVehicle: num 0.665 0.785 0.785 0.684 0.822 ...
## $ EndPerCapitaIncome : int 64040 166343 46499 56806 87451 83240 87451 113379 166343
27341 ...
## $ EndPctHouseholdsNoVehicle : num 0.545 0.684 0.806 0.584 0.823 ...
## $ TripDurationMinutes : num 6.98 7.56 11.48 17.15 21.32 ...
## $ DistanceMiles : num 0.849 0.989 1.334 2.35 1.825 ...
## $ StartStationName : chr "Atlantic Ave & Fort Greene P" "MacDougal St & Prince S
t" "MacDougal St & Prince St" "Warren St & Church St" ...
## $ StartStationLatitude : num 40.7 40.7 40.7 40.7 40.7 ...
## $ StartStationLongitude : num -74 -74 -74 -74 -74 ...
## $ StartNeighborhood : chr "Brooklyn" "Soho" "Soho" "Tribeca" ...
## $ EndStationName : chr "5 Ave & 3 St" "Murray St & West St" "E 9 St & Avenue C"
"Degraw St & Smith St" ...
## $ EndStationLatitude : num 40.7 40.7 40.7 40.7 40.7 ...
## $ EndStationLongitude : num -74 -74 -74 -74 -74 ...
## $ EndNeighborhood : chr "Park Slope" "Tribeca" "East Village" "Carroll Gardens"
...
```

```
colnames(citibike)
```

```
## [1] "DemandDate" "StartStationId"
## [3] "EndStationId" "DemandTime"
## [5] "Demand" "Temperature"
## [7] "DailySnowFall" "DailyPrecipitation"
## [9] "Customer_prop" "DayOfWeek"
## [11] "Month" "StartPerCapitaIncome"
## [13] "StartPctHouseholdsNoVehicle" "EndPerCapitaIncome"
## [15] "EndPctHouseholdsNoVehicle" "TripDurationMinutes"
## [17] "DistanceMiles" "StartStationName"
## [19] "StartStationLatitude" "StartStationLongitude"
## [21] "StartNeighborhood" "EndStationName"
## [23] "EndStationLatitude" "EndStationLongitude"
## [25] "EndNeighborhood"
```

```
head(citibike)
```

```

## DemandDate StartStationId EndStationId DemandTime Demand Temperature
## 1 6/1/2017 83 3368 evening 20 72.49123
## 2 6/1/2017 128 309 daytime 37 72.49123
## 3 6/1/2017 128 394 daytime 37 72.49123
## 4 6/1/2017 152 3329 evening 28 72.49123
## 5 6/1/2017 161 518 evening 79 72.49123
## 6 6/1/2017 168 402 daytime 53 72.49123
## DailySnowFall DailyPrecipitation Customer_prop DayOfWeek Month
## 1 0 NA 0.09649123 Thursday 6
## 2 0 NA 0.09649123 Thursday 6
## 3 0 NA 0.09649123 Thursday 6
## 4 0 NA 0.09649123 Thursday 6
## 5 0 NA 0.09649123 Thursday 6
## 6 0 NA 0.09649123 Thursday 6
## StartPerCapitaIncome StartPctHouseholdsNoVehicle EndPerCapitaIncome
## 1 61845 0.6653308 64040
## 2 115888 0.7852820 166343
## 3 115888 0.7852820 46499
## 4 166343 0.6842873 56806
## 5 79481 0.8217320 87451
## 6 113379 0.7938966 83240
## EndPctHouseholdsNoVehicle TripDurationMinutes DistanceMiles
## 1 0.5449030 6.983333 0.8488793
## 2 0.6842873 7.558333 0.9890215
## 3 0.8059011 11.483333 1.3339222
## 4 0.5844324 17.150000 2.3503307
## 5 0.8225607 21.316667 1.8247822
## 6 0.7635968 3.400000 0.2667268
## StartStationName StartStationLatitude StartStationLongitude
## 1 Atlantic Ave & Fort Greene P 40.68383 -73.97632
## 2 MacDougal St & Prince St 40.72710 -74.00297
## 3 MacDougal St & Prince St 40.72710 -74.00297
## 4 Warren St & Church St 40.71474 -74.00911
## 5 LaGuardia Pl & W 3 St 40.72917 -73.99810
## 6 W 18 St & 6 Ave 40.73971 -73.99456
## StartNeighborhood EndStationName EndStationLatitude EndStationLongitude
## 1 Brooklyn 5 Ave & 3 St 40.67282 -73.98352
## 2 Soho Murray St & West St 40.71498 -74.01301
## 3 Soho E 9 St & Avenue C 40.72521 -73.97769
## 4 Tribeca Degraw St & Smith St 40.68292 -73.99318
## 5 Greenwich Village E 39 St & 2 Ave 40.74780 -73.97344
## 6 Flatiron District Broadway & E 22 St 40.74034 -73.98955
## EndNeighborhood
## 1 Park Slope
## 2 Tribeca
## 3 East Village
## 4 Carroll Gardens
## 5 Murray Hill
## 6 Flatiron District

```

```
tail(citibike)
```

##	DemandDate	StartStationId	EndStationId	DemandTime	Demand	Temperature
## 31409	6/1/2018	528	267	evening	49	65.21429
## 31410	6/1/2018	540	336	evening	63	65.21429
## 31411	6/1/2018	3139	500	evening	15	65.21429
## 31412	6/1/2018	3158	3135	evening	60	65.21429
## 31413	6/1/2018	3164	261	evening	54	65.21429
## 31414	6/1/2018	3538	3459	evening	9	65.21429
##	DailySnowFall	DailyPrecipitation	Customer_prop	DayOfWeek	Month	
## 31409	0	0.1	NA	Friday	6	
## 31410	0	0.1	NA	Friday	6	
## 31411	0	0.1	NA	Friday	6	
## 31412	0	0.1	NA	Friday	6	
## 31413	0	0.1	NA	Friday	6	
## 31414	0	0.1	NA	Friday	6	
##	StartPerCapitaIncome	StartPctHouseholdsNoVehicle	EndPerCapitaIncome			
## 31409	87451	0.8225607	81964			
## 31410	87451	0.8225607	79481			
## 31411	116440	0.7085675	90151			
## 31412	111473	0.7372375	116440			
## 31413	111473	0.7372375	71656			
## 31414	63711	0.7378398	133504			
##	EndPctHouseholdsNoVehicle	TripDurationMinutes	DistanceMiles			
## 31409	0.8309120	20.12500	0.7869395			
## 31410	0.8217320	10.83333	1.2444371			
## 31411	0.8587258	14.01667	1.1828292			
## 31412	0.7085675	16.25000	1.3065352			
## 31413	0.6739155	37.15000	5.6859421			
## 31414	0.7416425	37.88333	3.1277485			
##	StartStationName	StartStationLatitude	StartStationLongitude			
## 31409	2 Ave & E 31 St	40.74291	-73.97706			
## 31410	Lexington Ave & E 29 St	40.74312	-73.98215			
## 31411	E 72 St & Park Ave	40.77118	-73.96409			
## 31412	W 63 St & Broadway	40.77164	-73.98261			
## 31413	Columbus Ave & W 72 St	40.77706	-73.97898			
## 31414	W 110 St & Amsterdam Ave	40.80269	-73.96295			
##	StartNeighborhood	EndStationName	EndStationLatitude			
## 31409	Kip Bay	Broadway & W 36 St	40.75098			
## 31410	Kip Bay Sullivan St & Washington Sq		40.73048			
## 31411	Upper East Side	Broadway & W 51 St	40.76229			
## 31412	Lincoln Square	E 75 St & 3 Ave	40.77113			
## 31413	Lincoln Square	Johnson St & Gold St	40.69475			
## 31414	Morningside Heights	E 53 St & 3 Ave	40.75763			
##	EndStationLongitude	EndNeighborhood				
## 31409	-73.98765	Garment District				
## 31410	-73.99906	Greenwich Village				
## 31411	-73.98336	Theater District-Times Square				
## 31412	-73.95772	Upper East Side				
## 31413	-73.98362	Downtown Brooklyn				
## 31414	-73.96931	Turtle Bay				

Correct the type of R objects

```
# default takes two values and is a factor
```

```
citibikeselect <- select(citibike, DemandDate, StartStationId, EndStationId, DemandTime, Demand,
Temperature, DailyPrecipitation, Customer_prop, DayOfWeek, Month, StartPerCapitaIncome, StartPct
HouseholdsNoVehicle, TripDurationMinutes, DistanceMiles, StartStationName, StartNeighborhood, End
Neighborhood)
```

```
str(citibikeselect)
```

```
## 'data.frame': 31414 obs. of 17 variables:
## $ DemandDate : chr "6/1/2017" "6/1/2017" "6/1/2017" "6/1/2017" ...
## $ StartStationId : int 83 128 128 152 161 168 173 223 229 239 ...
## $ EndStationId : int 3368 309 394 3329 518 402 526 168 309 421 ...
## $ DemandTime : chr "evening" "daytime" "daytime" "evening" ...
## $ Demand : int 20 37 37 28 79 53 47 30 48 41 ...
## $ Temperature : num 72.5 72.5 72.5 72.5 72.5 ...
## $ DailyPrecipitation : num NA NA NA NA NA NA NA NA NA ...
## $ Customer_prop : num 0.0965 0.0965 0.0965 0.0965 0.0965 ...
## $ DayOfWeek : chr "Thursday" "Thursday" "Thursday" "Thursday" ...
## $ Month : int 6 6 6 6 6 6 6 6 6 6 ...
## $ StartPerCapitaIncome : int 61845 115888 115888 166343 79481 113379 90151 113379 851
99 71656 ...
## $ StartPctHouseholdsNoVehicle: num 0.665 0.785 0.785 0.684 0.822 ...
## $ TripDurationMinutes : num 6.98 7.56 11.48 17.15 21.32 ...
## $ DistanceMiles : num 0.849 0.989 1.334 2.35 1.825 ...
## $ StartStationName : chr "Atlantic Ave & Fort Greene P" "MacDougal St & Prince S
t" "MacDougal St & Prince St" "Warren St & Church St" ...
## $ StartNeighborhood : chr "Brooklyn" "Soho" "Soho" "Tribeca" ...
## $ EndNeighborhood : chr "Park Slope" "Tribeca" "East Village" "Carroll Gardens"
...
```

```
citibikeselect <- citibikeselect %>%
  mutate(
    Temperature = ifelse(is.na(Temperature), mean(Temperature, na.rm = TRUE), Temperature),
    DailyPrecipitation = ifelse(is.na(DailyPrecipitation), mean(DailyPrecipitation, na.rm = TRUE), DailyPrecipitation),
    Customer_prop = ifelse(is.na(Customer_prop), mean(Customer_prop, na.rm = TRUE), Customer_prop),
    StartPerCapitaIncome = ifelse(is.na(StartPerCapitaIncome), mean(StartPerCapitaIncome, na.rm = TRUE), StartPerCapitaIncome),
    StartPctHouseholdsNoVehicle = ifelse(is.na(StartPctHouseholdsNoVehicle), mean(StartPctHouseholdsNoVehicle, na.rm = TRUE), StartPctHouseholdsNoVehicle))

citibikeselect$StartStationId <- as.factor(citibikeselect$StartStationId)
citibikeselect$EndStationId <- as.factor(citibikeselect$EndStationId)
citibikeselect$DemandTime <- as.factor(citibikeselect$DemandTime)
citibikeselect$DayOfWeek <- as.factor(citibikeselect$DayOfWeek)
citibikeselect$Month <- as.factor(citibikeselect$Month)
citibikeselect$StartNeighborhood <- as.factor(citibikeselect$StartNeighborhood)
citibikeselect$EndNeighborhood <- as.factor(citibikeselect$EndNeighborhood)

citibikeselect$DemandDate <- as.Date(citibikeselect$DemandDate, format = "%m/%d/%y")
```

Run descriptive statistics

Compute descriptive stats (mean and standard deviation) for balance and income and frequency distribution for default and student. Feel free to use dplyr functions if needed.

```
# Descriptive stats for continuous variables: mean, median, min, max, sd
# balance

summary(citibikeselect)
```

```

##      DemandDate      StartStationId  EndStationId      DemandTime
##  Min.   :2020-01-01    519      : 305    519      : 312    daytime:12139
##  1st Qu.:2020-05-04    426      : 220    426      : 223    evening:19275
##  Median :2020-07-19    435      : 217    435      : 215
##  Mean   :2020-07-13    497      : 208    497      : 215
##  3rd Qu.:2020-09-28    402      : 202    402      : 201
##  Max.   :2020-12-31    285      : 190    514      : 198
##
##      (Other):30072    (Other):30050
##      Demand      Temperature      DailyPrecipitation Customer_prop
##  Min.   : 1.00    Min.   :11.11    Min.   :0.00000    Min.   :0.009615
##  1st Qu.: 21.00    1st Qu.:49.99    1st Qu.:0.00000    1st Qu.:0.054054
##  Median : 40.00    Median :66.40    Median :0.01591    Median :0.082192
##  Mean   : 45.36    Mean   :62.58    Mean   :0.07403    Mean   :0.102393
##  3rd Qu.: 60.00    3rd Qu.:75.59    3rd Qu.:0.07403    3rd Qu.:0.125000
##  Max.   :294.00    Max.   :87.64    Max.   :2.65977    Max.   :0.377358
##
##      DayOfWeek      Month      StartPerCapitaIncome
##  Friday   :4853    5      : 3554    Min.   : 19071
##  Monday   :4222    10     : 3492    1st Qu.: 63711
##  Saturday :4001    9      : 3408    Median : 85199
##  Sunday   :3525    8      : 3300    Mean   : 82565
##  Thursday :5086    7      : 3201    3rd Qu.:106735
##  Tuesday  :4798    6      : 3051    Max.   :166343
##  Wednesday:4929    (Other):11408
##  StartPctHouseholdsNoVehicle TripDurationMinutes DistanceMiles
##  Min.   :0.4864                Min.   : 1.017    Min.   :0.0000
##  1st Qu.:0.7162                1st Qu.: 6.314    1st Qu.:0.5192
##  Median :0.7902                Median :10.283    Median :0.8628
##  Mean   :0.7630                Mean   :12.977    Mean   :1.1232
##  3rd Qu.:0.8146                3rd Qu.:17.333    3rd Qu.:1.4565
##  Max.   :0.8624                Max.   :59.633    Max.   :7.3879
##
##  StartStationName      StartNeighborhood      EndNeighborhood
##  Length:31414          Chelsea      : 2548    Chelsea      : 2529
##  Class :character      Hells Kitchen : 1954    Hells Kitchen : 2011
##  Mode  :character      Greenwich Village: 1512    Greenwich Village: 1485
##
##      East Village      : 1428    East Village      : 1420
##
##      Upper West Side : 1155    Greenpoint      : 1160
##
##      Greenpoint      : 1144    Lower East Side : 1122
##
##      (Other)      :21673    (Other)      :21687

```

```

cor(select(citibikeselect, Demand, Temperature, DailyPrecipitation, Customer_prop, StartPerCapitaIncome, StartPctHouseholdsNoVehicle, TripDurationMinutes, DistanceMiles))

```

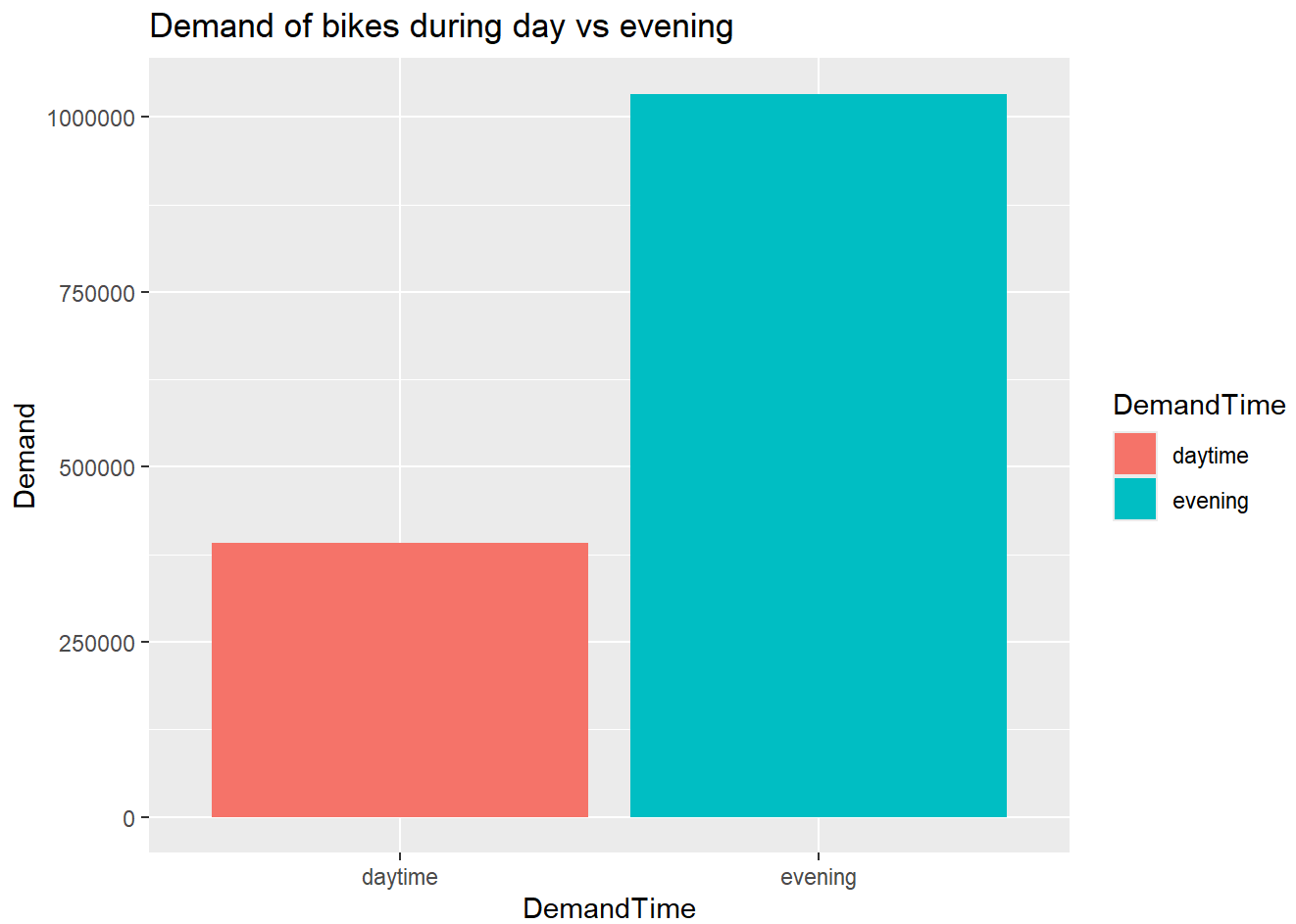
```
##                                Demand Temperature DailyPrecipitation
## Demand                        1.0000000000 0.020590827 0.0007565156
## Temperature                   0.0205908274 1.000000000 0.0034445493
## DailyPrecipitation            0.0007565156 0.003444549 1.0000000000
## Customer_prop                 -0.0160536714 0.357145121 -0.0117761993
## StartPerCapitaIncome          0.3201634536 0.035487609 -0.0068886615
## StartPctHouseholdsNoVehicle   0.3873014561 0.002821490 -0.0097298258
## TripDurationMinutes           0.0023209146 0.109468032 -0.0136923355
## DistanceMiles                 0.0068774994 0.072224511 -0.0141215998
##                                Customer_prop StartPerCapitaIncome
## Demand                        -0.01605367 0.320163454
## Temperature                   0.35714512 0.035487609
## DailyPrecipitation            -0.01177620 -0.006888662
## Customer_prop                 1.000000000 -0.016398167
## StartPerCapitaIncome          -0.01639817 1.000000000
## StartPctHouseholdsNoVehicle   -0.05453646 0.129833238
## TripDurationMinutes           0.13350613 0.043992015
## DistanceMiles                 0.03981854 0.042154653
##                                StartPctHouseholdsNoVehicle TripDurationMinutes
## Demand                        0.387301456 0.002320915
## Temperature                   0.002821490 0.109468032
## DailyPrecipitation            -0.009729826 -0.013692335
## Customer_prop                 -0.054536458 0.133506129
## StartPerCapitaIncome          0.129833238 0.043992015
## StartPctHouseholdsNoVehicle   1.000000000 0.004012768
## TripDurationMinutes           0.004012768 1.000000000
## DistanceMiles                 0.025525009 0.761967809
##                                DistanceMiles
## Demand                        0.006877499
## Temperature                   0.072224511
## DailyPrecipitation            -0.014121600
## Customer_prop                 0.039818542
## StartPerCapitaIncome          0.042154653
## StartPctHouseholdsNoVehicle   0.025525009
## TripDurationMinutes           0.761967809
## DistanceMiles                 1.000000000
```

Visualization

Use ggplot() from ggplot2 package to create different plots of variables.

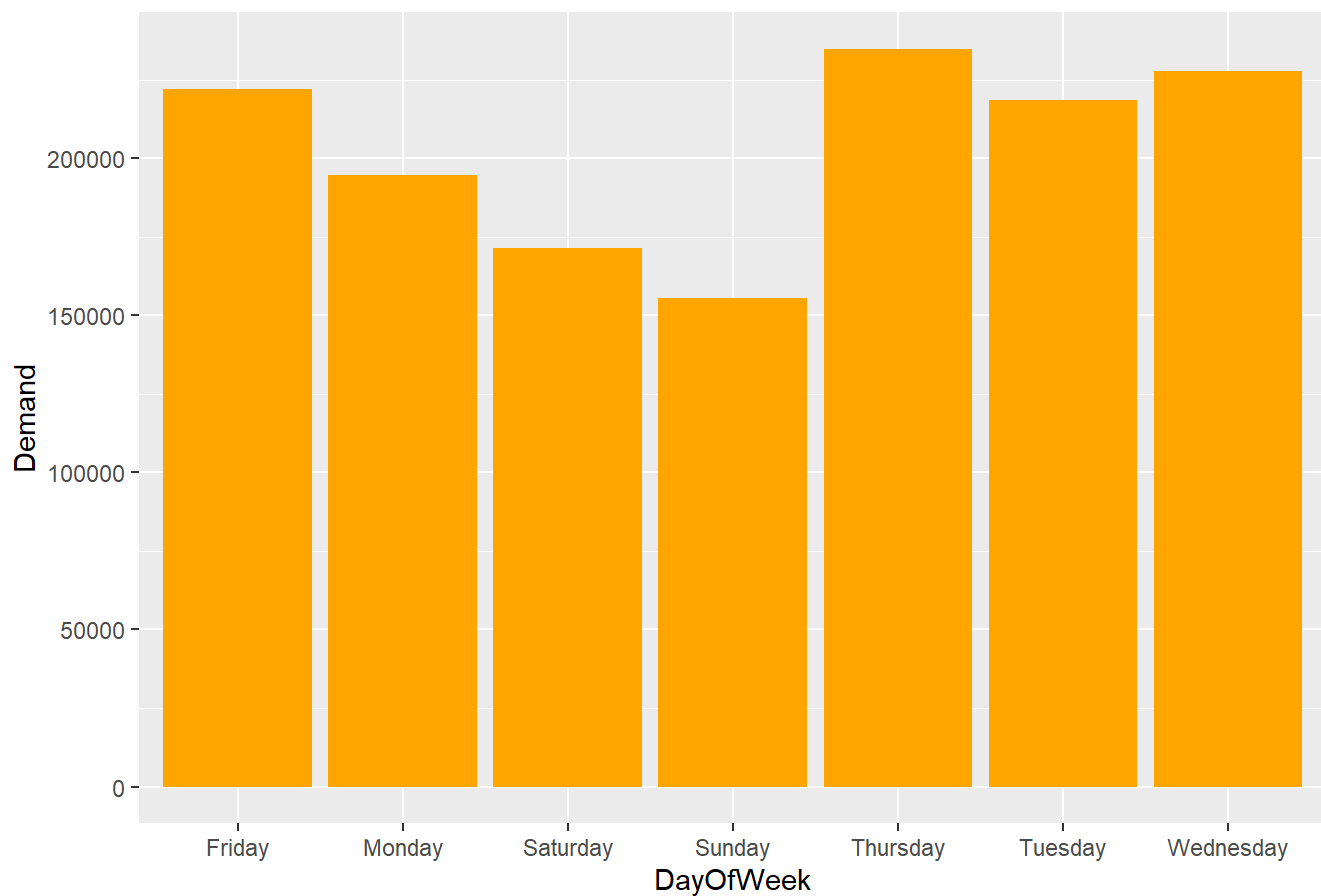
```
#Plotting of Demand against various parameters to understand basic trends and correlations

#Comparison of Demand during Daytime and Evening using bar chart
ggplot(citibikeselect, aes(x = DemandTime, y = Demand, fill=DemandTime)) +
  geom_bar(stat="identity")+
  ggtitle("Demand of bikes during day vs evening") +
  xlab("DemandTime") + ylab("Demand")
```

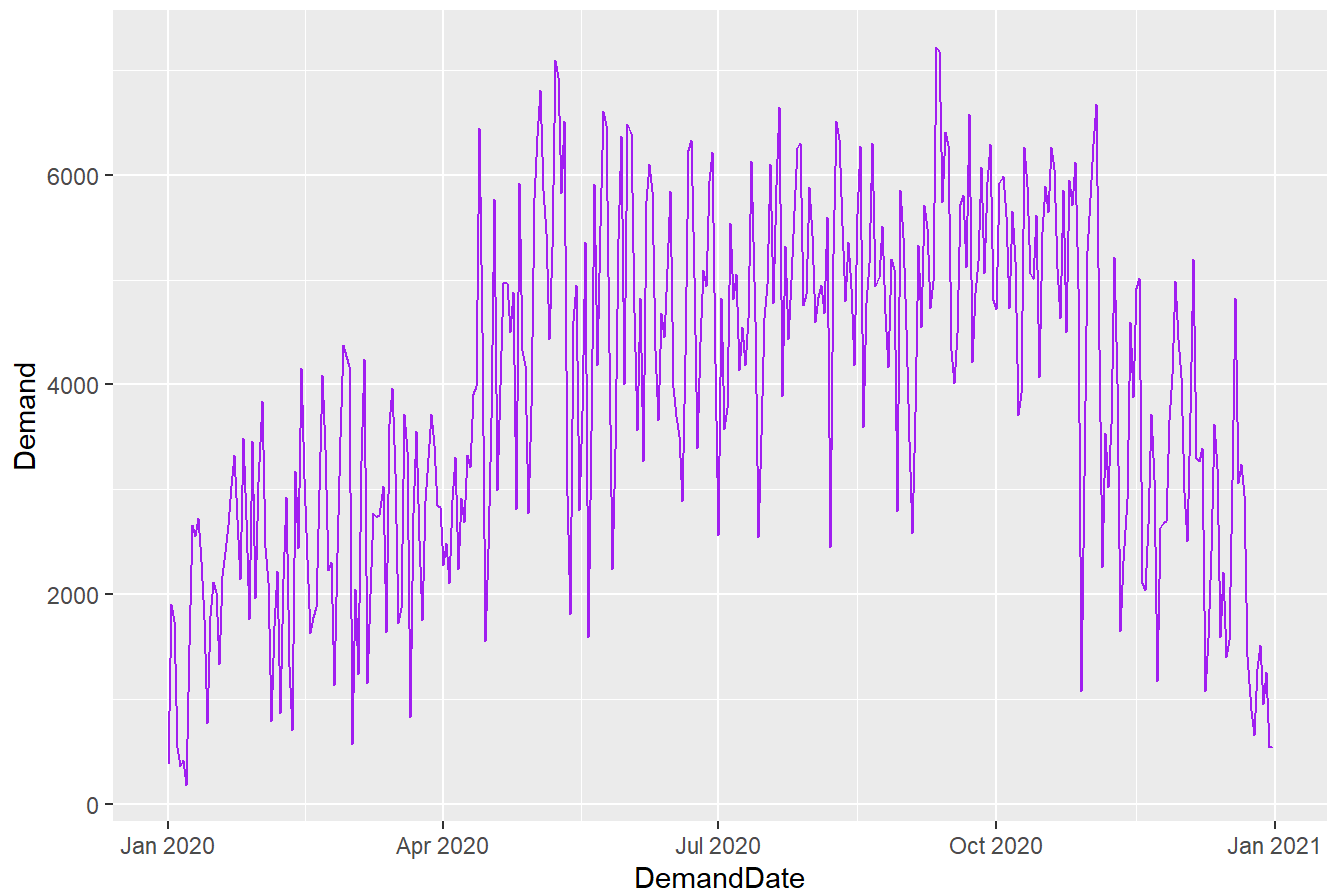
```
#DayofWeek vs Demand barchart
ggplot(citibikeselect, aes(x = DayOfWeek, y = Demand)) +
  geom_bar(stat="identity", fill = "orange")+
  ggtitle("Demand of bikes during day vs evening") +
  xlab("DayOfWeek") + ylab("Demand")
```

Demand of bikes during day vs evening



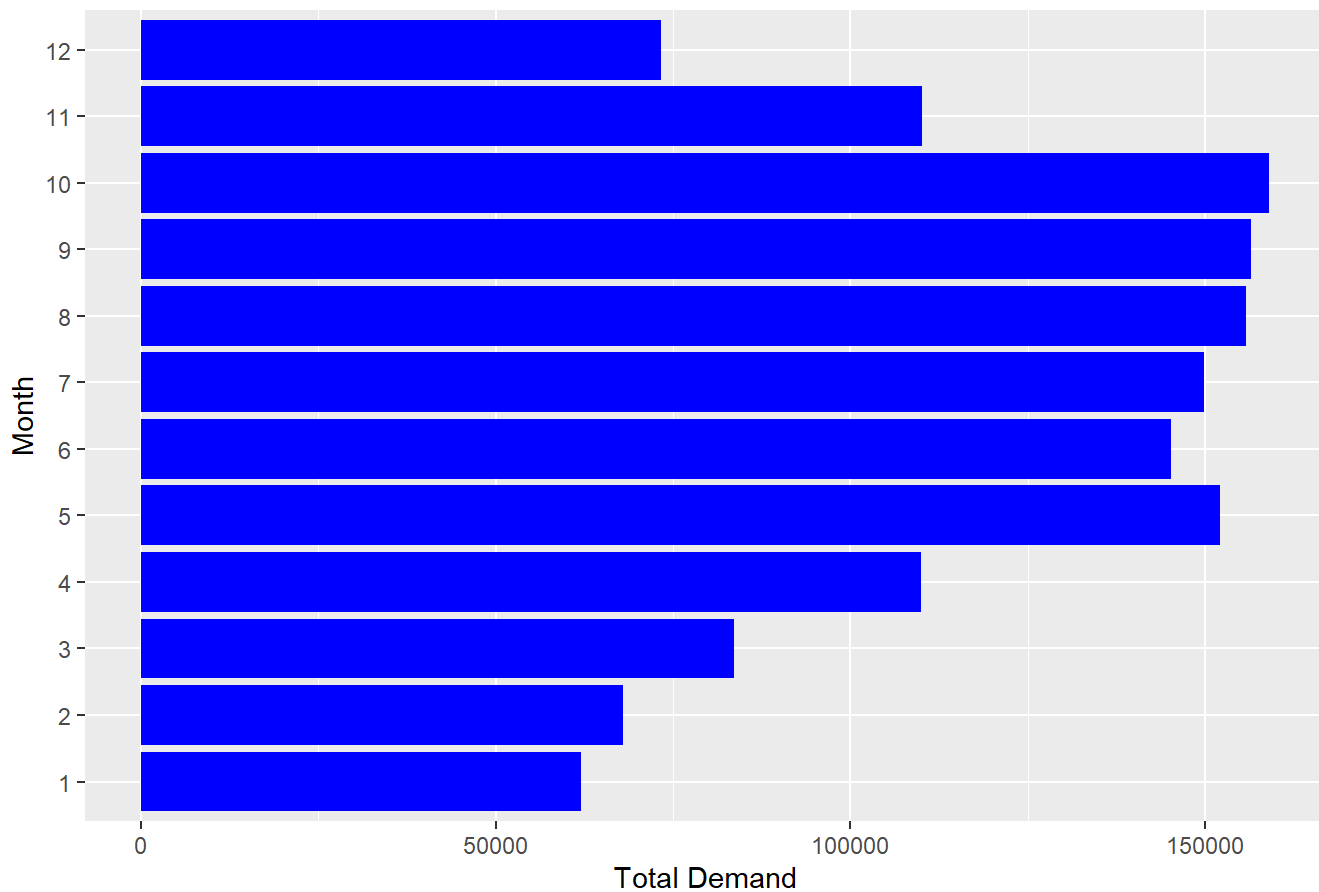
```
#Demand of Citibikes over the course of one year(range of our dataset)
ggplot(citibikeselect, aes(x = DemandDate, y = Demand)) +
  geom_line(stat = "summary", fun = "sum",color = "purple") +
  ggtitle("Demand over time") + xlab("DemandDate")+ ylab("Demand")
```

Demand over time



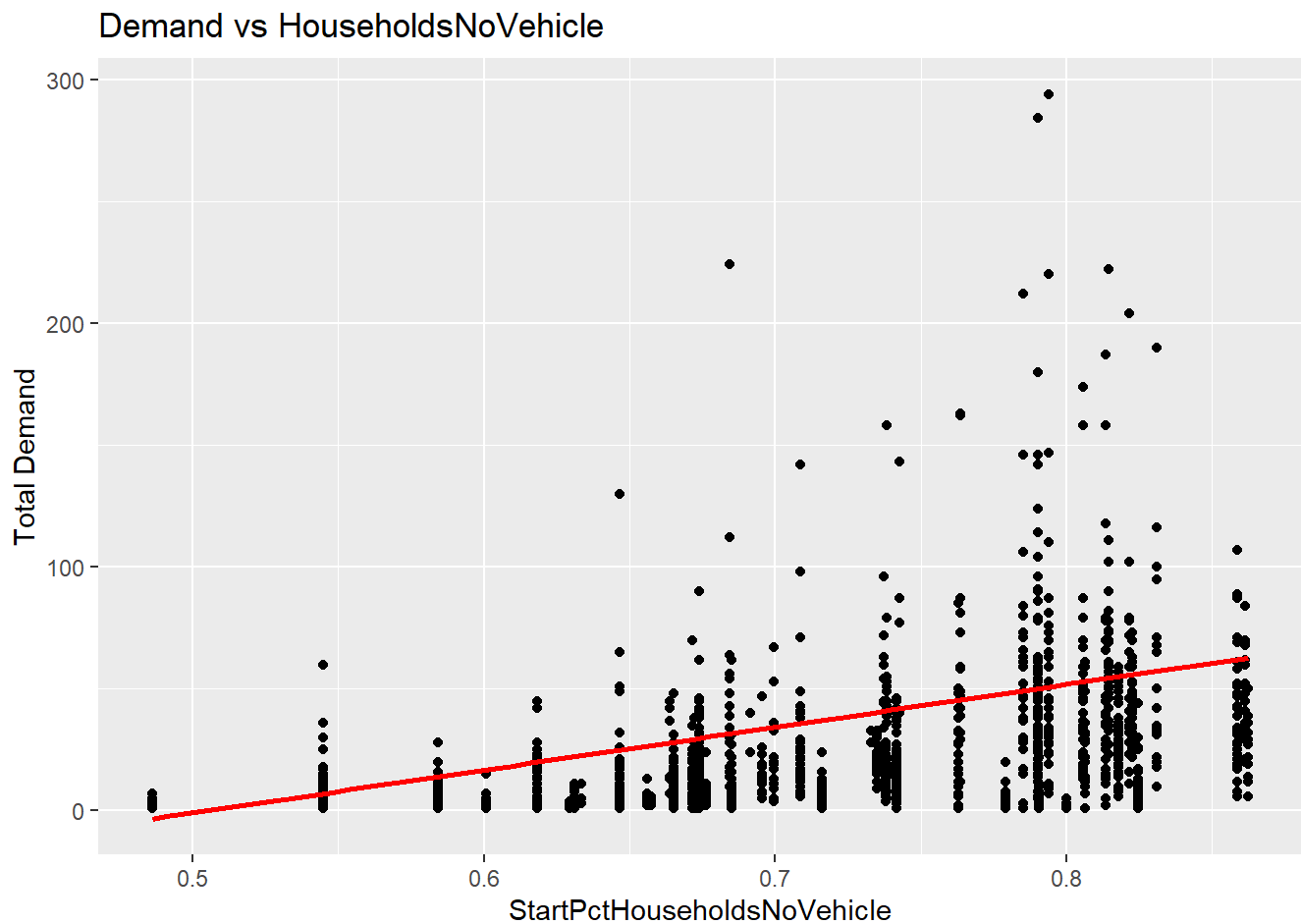
```
#Monthly demand of Citibikes  
ggplot(citibikeselect, aes(x = Demand , y = Month)) +  
  geom_bar(stat = "summary", fun = "sum", fill = "blue") +  
  labs(title = "Demand over Month", x = "Total Demand", y = "Month")
```

Demand over Month



```
#Scatterplot to understand correlation between StartPctHouseholdsNoVehicle and Demand
ggplot(citibikeselect, aes(x = StartPctHouseholdsNoVehicle, y = Demand)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(title = "Demand vs HouseholdsNoVehicle", x = "StartPctHouseholdsNoVehicle", y = "Total Demand")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
#Scatterplot to understand correlation between StartPerCapitaIncome and Demand  
ggplot(citibikeselect, mapping = aes(x = StartPerCapitaIncome, y = Demand)) +  
  geom_point()+  
  geom_smooth(method = "lm", se = FALSE, color = "red") +  
  ggtitle("Demand vs Income") + xlab("income") + ylab("Demand")
```

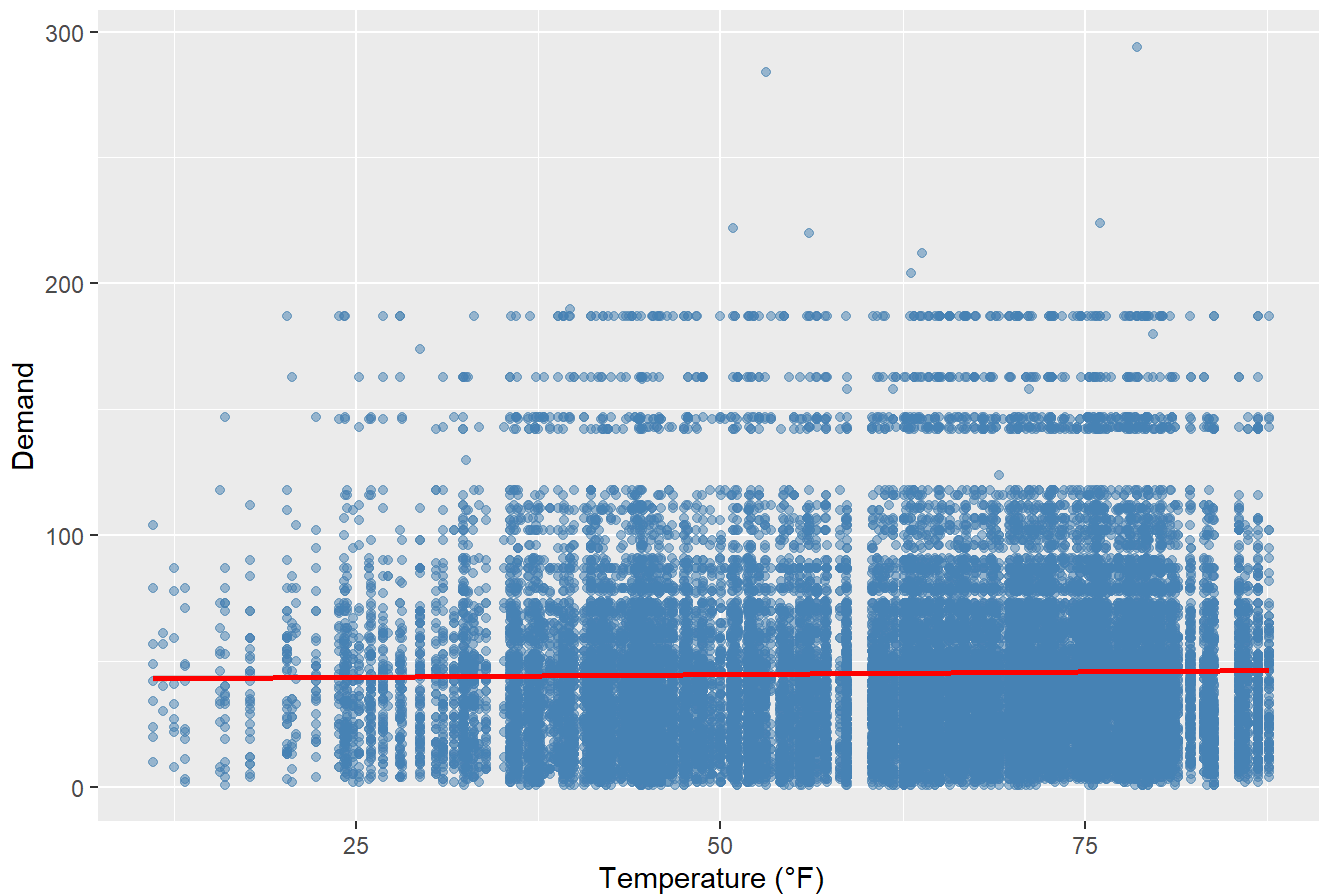
```
## `geom_smooth()` using formula = 'y ~ x'
```



```
#Scatterplot to understand correlation between Temperature and Demand
ggplot(citibikeselect, aes(x = Temperature, y = Demand)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", color = "red") +
  labs(title = "Temperature vs Bike Demand",
       x = "Temperature (°F)", y = "Demand")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Temperature vs Bike Demand



Using the following filter, group_by, and head functions to determine the stations with the highest demand for daytime and evening.

```
daytime_stations <- filter(citibikeselect, DemandTime == "daytime")
top_daytime <- daytime_stations %>%
  group_by(StartStationId, StartStationName) %>%
  summarise(daysum = sum(Demand))
```

`summarise()` has grouped output by 'StartStationId'. You can override using
the `.groups` argument.

```
head_day <- print(head(arrange(top_daytime, desc(daysum))))
```

```
## # A tibble: 6 × 3
## # Groups:   StartStationId [6]
##   StartStationId StartStationName    daysum
##   <fct>          <chr>              <int>
## 1 519            Pershing Square North  13924
## 2 3255           8 Ave & W 31 St    10404
## 3 432            E 7 St & Avenue A    7656
## 4 490            8 Ave & W 33 St    6084
## 5 426            West St & Chambers St    5929
## 6 514            12 Ave & W 40 St    5041
```

```
evening_stations <- filter(citibikeselect, DemandTime == "evening")
top_evening <- evening_stations %>%
  group_by(StartStationId, StartStationName) %>%
  summarise(eveningsum = sum(Demand))
```

```
## `summarise()` has grouped output by 'StartStationId'. You can override using
## the `.groups` argument.
```

```
head_evening <- print(head(arrange(top_evening, desc(eveningsum))))
```

```
## # A tibble: 6 × 3
## # Groups:   StartStationId [6]
##   StartStationId StartStationName     eveningsum
##   <fct>          <chr>              <int>
## 1 519            Pershing Square North    34969
## 2 402            Broadway & E 22 St      26569
## 3 435            W 21 St & 6 Ave        21756
## 4 497            E 17 St & Broadway        21316
## 5 426            West St & Chambers St      20449
## 6 285            Broadway & E 14 St        20306
```

```
reg_citibike <- lm(Demand ~ DemandTime + DayOfWeek + Month, data = citibikeselect)
summary(reg_citibike)
```



```
##
## Call:
## lm(formula = Demand ~ DemandTime + DayOfWeek + Month, data = citibikeselect)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -55.695 -20.439  -4.747  14.716 239.124
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    32.49832     0.95845   33.907 < 2e-16 ***
## DemandTimeevening 21.47373     0.35846   59.906 < 2e-16 ***
## DayOfWeekMonday    0.50127     0.65024    0.771  0.4408
## DayOfWeekSaturday -3.79504     0.66012  -5.749 9.06e-09 ***
## DayOfWeekSunday   -3.19296     0.68539  -4.659 3.20e-06 ***
## DayOfWeekThursday  0.80378     0.62022    1.296  0.1950
## DayOfWeekTuesday   0.64913     0.62920    1.032  0.3022
## DayOfWeekWednesday 0.88023     0.62530    1.408  0.1592
## Month2            -0.37300     1.14661   -0.325  0.7450
## Month3            -0.88497     1.09539   -0.808  0.4192
## Month4            -0.90720     1.03776   -0.874  0.3820
## Month5            -2.52127     0.98116  -2.570  0.0102 *
## Month6             1.84276     1.00464    1.834  0.0666 .
## Month7             0.90350     0.99705    0.906  0.3648
## Month8             1.59241     0.99224    1.605  0.1085
## Month9             0.78529     0.98733    0.795  0.4264
## Month10            0.14167     0.98333    0.144  0.8854
## Month11            -0.05916     1.04105   -0.057  0.9547
## Month12            -0.22553     1.12698   -0.200  0.8414
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.85 on 31395 degrees of freedom
## Multiple R-squared:  0.1057, Adjusted R-squared:  0.1052
## F-statistic: 206.1 on 18 and 31395 DF, p-value: < 2.2e-16
```

```
# Predicting Demand for 08/18/2018 (Saturday).
```

```
# Daytime Predictions
```

```
# Origin Station: Pershing Square North (519)
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "519", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##      1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "519", EndStationId =
"3255", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "519", EndStationId =
"432", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "519", EndStationId =
"490", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "519", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
# Origin Station: 8 Ave & W 31 St (3255)
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "3255", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "3255", EndStationId =
"3255", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "3255", EndStationId =
"432", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "3255", EndStationId =
"490", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "3255", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
# Origin Station: E 7 St & Avenue A (432)
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "432", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "432", EndStationId =
"3255", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "432", EndStationId =
"432", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "432", EndStationId =
"490", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "432", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
# Origin Station: 8 Ave & W 33 St (490)
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "490", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "490", EndStationId =
"3255", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "490", EndStationId =
"432", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "490", EndStationId =
"490", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "490", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
# Origin Station: West St & Chambers St (426)
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "426", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "426", EndStationId =
"3255", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "426", EndStationId =
"432", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "426", EndStationId =
"490", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
predict(reg_citibike, data.frame(DemandTime = "daytime", StartStationId = "426", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 30.29568
```

```
# Evening Predictions
# Origin Station: Pershing Square North (519)
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "519", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "519", EndStationId =
"402", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "519", EndStationId =
"435", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "519", EndStationId =
"497", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "519", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
# Origin Station: Broadway & E 22 St (402)
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "402", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "402", EndStationId =
"402", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "402", EndStationId =
"435", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "402", EndStationId =
"497", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "402", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
# Origin Station: W 21 St & 6 Ave (435)
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "435", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "435", EndStationId =
"402", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "435", EndStationId =
"435", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "435", EndStationId =
"497", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "435", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
# Origin Station: E 17 St & Broadway (497)
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "497", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "497", EndStationId =
"402", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "497", EndStationId =
"435", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "497", EndStationId =
"497", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "497", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
# Origin Station: West St & Chambers St (426)
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "426", EndStationId =
"519", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "426", EndStationId =
"402", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "426", EndStationId =
"435", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "426", EndStationId =
"497", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```

```
predict(reg_citibike, data.frame(DemandTime = "evening", StartStationId = "426", EndStationId =
"426", DayOfWeek = "Saturday", Month = "8"))
```

```
##          1
## 51.76941
```